Knowledge Organiser
Year 8: Human Body - Diffusion

What is diffusion?
- All particles **are randomly vibrating**, the particles in a liquid or gas are also moving around
- Diffusion is the movement of a substance from an area of **high concentration** to an area of l**ow concentration**
- It is an important process for living things, it is how substances move in and out of cells



The blue dye is concentrated in one area

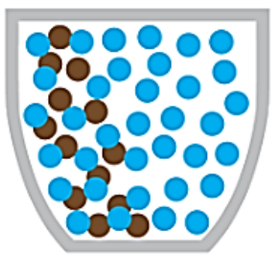
The dye particles have moved and **spread out** to the areas of lower concentration

How does diffusion happen?
- The particles in gases and liquids can move around and **collide** with each other randomly
- If there are more particles concentrated in one area they are **more likely to collide**
- This means that over time particles **spread out** until they have a consistent concentration

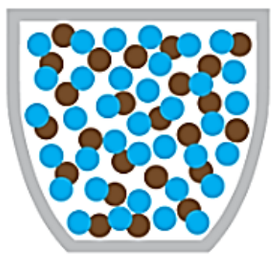
A particle diagram showing how coffee mixes with water through diffusion



1. Highly concentrated coffee molecules enter the cup of hot water.



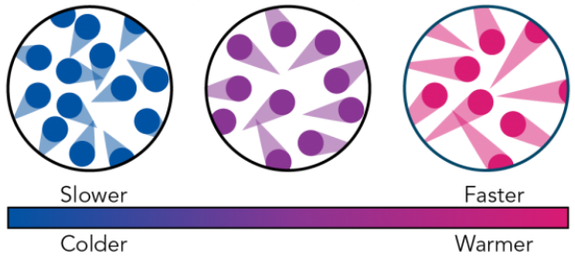
2. Coffee molecules begin to spread out in between the water molecules.



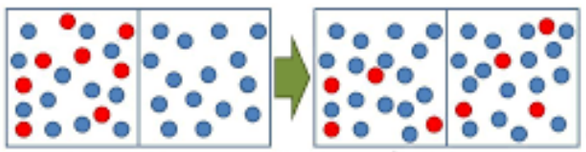
3. Coffee molecules are now in a lower concentration than they started in.

Diffusion doesn't always happen at the same rate:
- Although diffusion happens in all substances with moving particles, it doesn't always happen at the same rate
- Factors like **temperature, concentration difference** and **type of substance** can affect the rate of diffusion

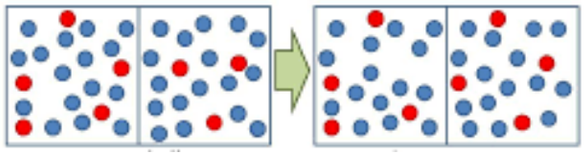
Temperature: At higher temperatures particles have more **kinetic energy**. This means the particles are moving around more quickly, so spread out more easily.



Concentration: If there is a bigger **difference in concentration** between the high and low concentration areas, particles are more likely to collide and spread out.



Large difference in concentration: Faster rate of diffusion



Small difference in concentration: Slower rate of diffusion

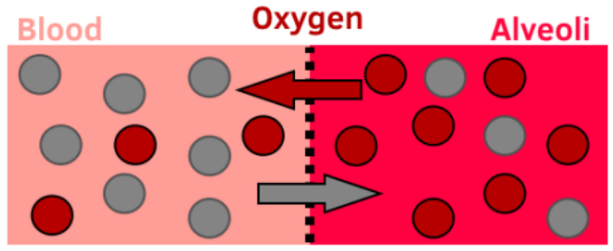
Type of substance: Substances will diffuse more quickly through gases than liquids. This is because the particles in a gas are more spread out and move faster.



Think…how long does it take for a bad smell to spread through a room? It happens very quickly because the particles in a gas move quickly!

Diffusion in biological processes
- Diffusion is one of the processes that moves substances into and out of our cells
- If a substance is higher in concentration outside of a cell, it will move across the **cell membrane** through diffusion

Diffusion in the lungs
- When we breath in the **alveoli** (air sacs) in our lungs fill with oxygen molecules
- This means there is a higher concentration of oxygen in the **alveoli** than in the blood which travels around them
- The oxygen will move through the lining of the alveoli and into the lower concentration area in the blood

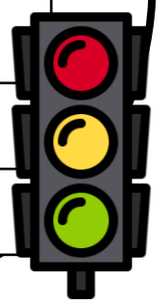


Remember: oxygen is converted into **carbon dioxide** during **aerobic respiration**. The carbon dioxide builds up in our blood until it diffuses back into the alveoli.

Passive vs. Active transport?
- Diffusion is a type of **"passive**" transport. This means it happens without the need for extra energy
- Different processes use "**active"** transport which uses energy to transport substances from a low to high concentration area

RAG Yourself!

Learning Outcomes;	RAG
Describe how substances move through diffusion	
State the factors which affect the rate of diffusion	
Describe what is meant by rate of diffusion	
Explain why the rate of diffusion is quicker at higher temperatures	
Explain why the rate of diffusion is quicker when there is a higher concentration gradient	
Describe how diffusion helps us to breathe and respire	



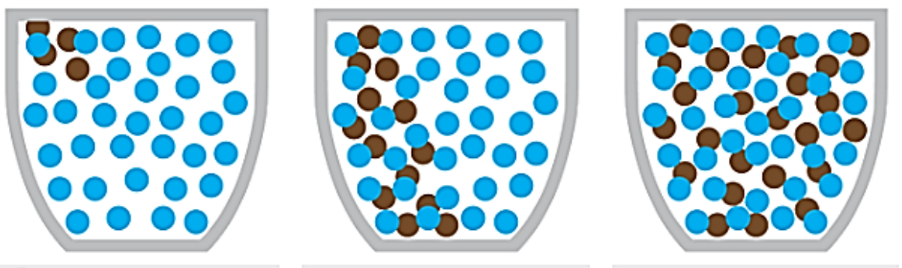
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What is diffusion?



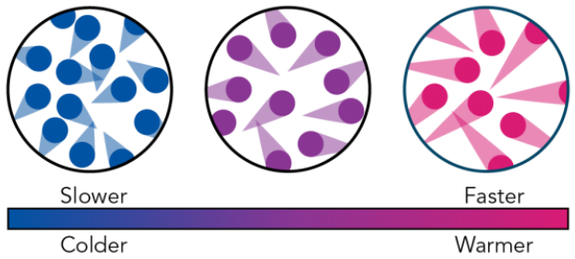
How does diffusion happen?

A particle diagram showing how coffee mixes with water through diffusion



Diffusion doesn't always happen at the same rate:

Temperature:



Concentration:

Large difference in concentration: Faster rate of diffusion

Small difference in concentration: Slower rate of diffusion

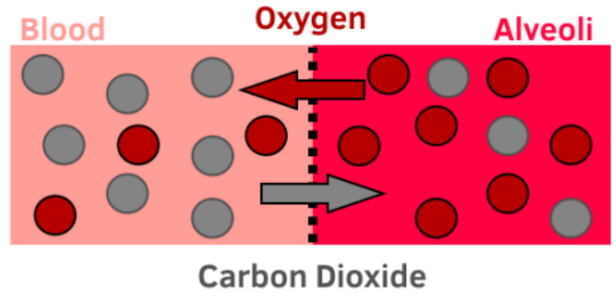
Type of substance:



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Diffusion in biological processes

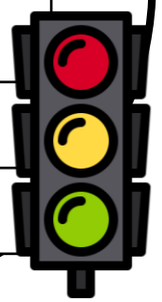
Diffusion in the lungs



Passive vs. Active transport?

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Describe how diffusion helps us to breathe and respire	



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